Introducción a la Electrónica

Unidad 7

El Amplificador Operacional de Tensión (AOV)

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TP 10

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5. A partir de $t = 0$ se aplican e_1, e_2 y e_3.
   Calcular y graficar:
   a) e_o.
   b) v_{id} del AOV.

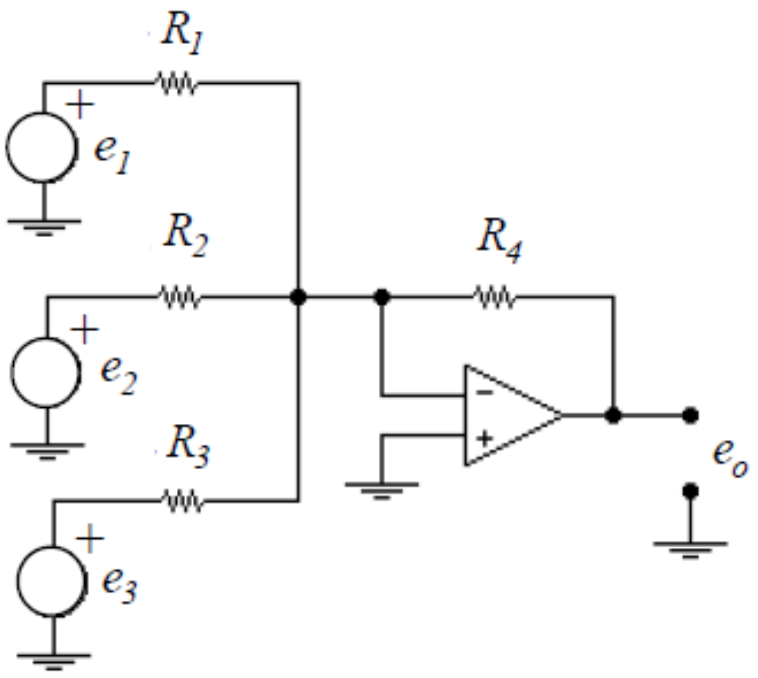
$$e_1 = 1V.sen^2 \omega t$$

$$e_1 = 1V.cos^2 \omega t$$

$$e_3 = 0{,}5V$$

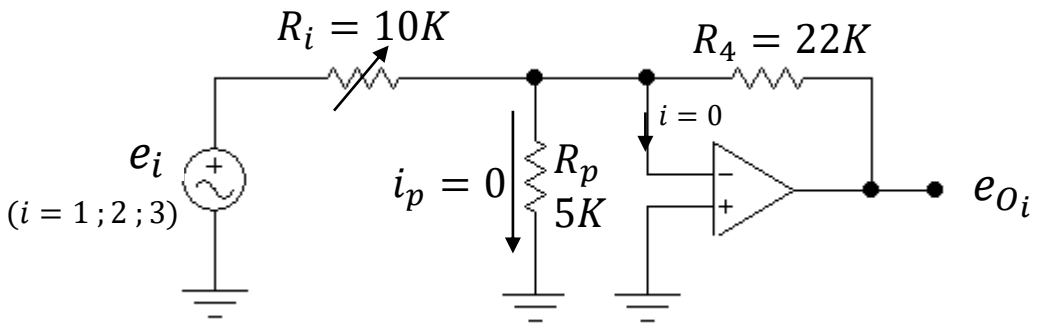
$$R_1 = R_2 = R_3 = 10K$$

$$R_4 = 22K$$



Aplicando superposición:

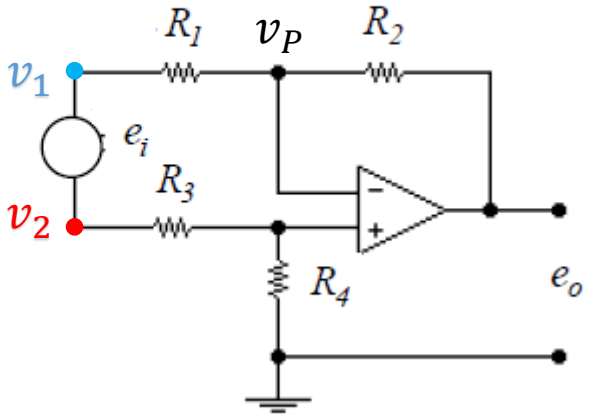
$$e_{O_i} = -\frac{R_4}{R_i} \,.\, e_i$$



$$e_O = -\sum_{i=1}^{3} \frac{R_4}{R_i} \,.\, e_i = -\frac{R_4}{R_1} \,.\, (e_1 + e_2 + e_3) = -k.\sum e_i \quad \Rightarrow \quad Amplificador\ Sumador\ (e\ Inversor)$$

$$e_O = -2{,}2\,.\,[1V.sen^2(\omega t) + 1V.cos^2(\omega t) + 0{,}5V] = -3{,}3V$$

6. En éste caso, ¿cuál es la relación entre salida y entrada?



Si planteamos la superposición de v_1 y v_2 podremos escribir que:

$$e_O = e_{O_1} + e_{O_2}$$

$a)$ $v_2 = 0 \quad \Rightarrow \quad v_{i+} = 0$

$$v_P = v_{i-} = v_{i+} = 0$$

$$e_{O_1} = -\frac{R_2}{R_1}.v_1$$

$b)$ $v_1 = 0 \quad \Rightarrow \quad v_{i+} = v_2.\left(\frac{R_4}{R_3 + R_4}\right)$

$$v_P = v_2.\left(\frac{R_4}{R_3 + R_4}\right)$$

$$e_{O_2} = \left(1 + \frac{R_2}{R_1}\right).v_2.\left(\frac{R_4}{R_3 + R_4}\right) =$$

$$= v_2.\left(\frac{R_1 + R_2}{R_1}\right).\left(\frac{R_4}{R_3 + R_4}\right)$$

$c)$ si se cumple que:

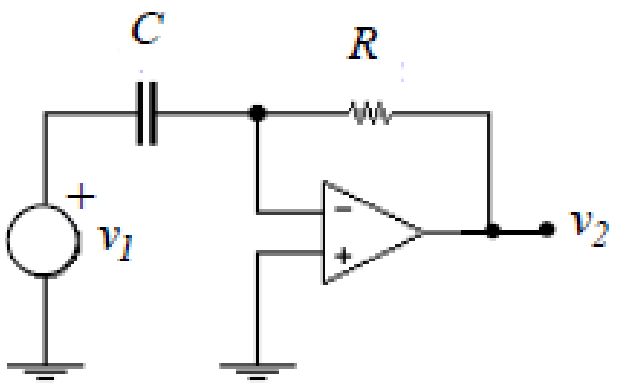
$$\frac{R_4}{R_3} = \frac{R_2}{R_1}$$

se verifica que:

$$e_O = \left(\frac{R_2}{R_1}\right).(v_2 - v_1) = \left(\frac{R_2}{R_1}\right).e_i$$

Amplificador Restador (Diferencial)

7. a) ¿Cómo se denomina el siguiente circuito y por qué?
b) Realizar la gráfica de $v_2(t)$



$$v_1 = 1\ V.\ sen(2000.\pi.t)$$
$$R = 1K$$
$$C = 1\mu F$$

$$i_R = i_C = C.\frac{dv_C}{dt} = C.\frac{dv_1}{dt}$$

$$v_2 = -i_R.R = -RC.\frac{dv_1}{dt} = k.\frac{dv_1}{dt} \quad \Rightarrow \quad Amplificador\ Derivador\ ó\ Diferenciador\ e\ Inversor$$

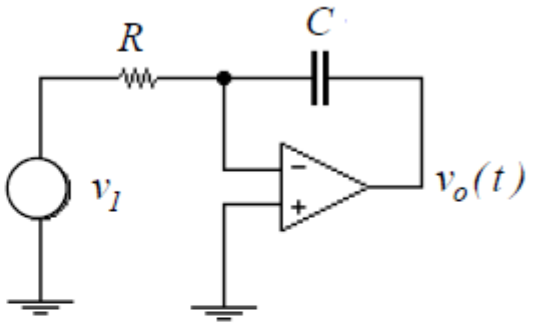
8. En el siguiente circuito a partir de $t = 0$ se aplica un escalón de $1V$ a la entrada.

Graficar:

a) $v_1(t)$

b) $v_o(t)$

c) Indicar la polaridad de la tensión en C.

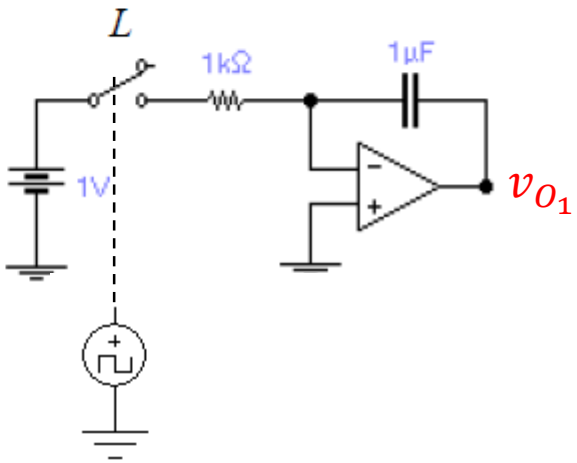


Datos:

$v_1 = 1\ V.\ \mu(t)$

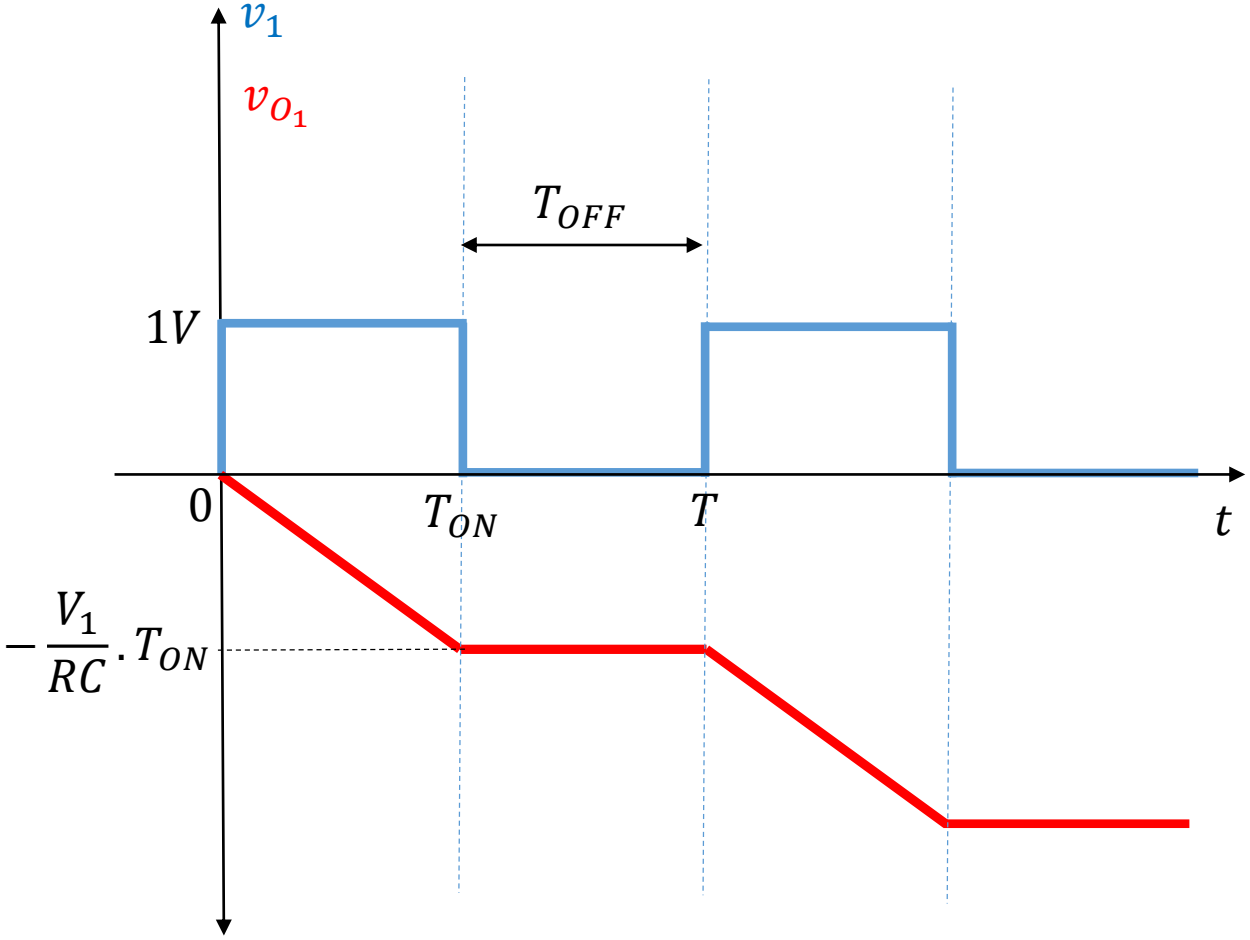
$R = 1K$

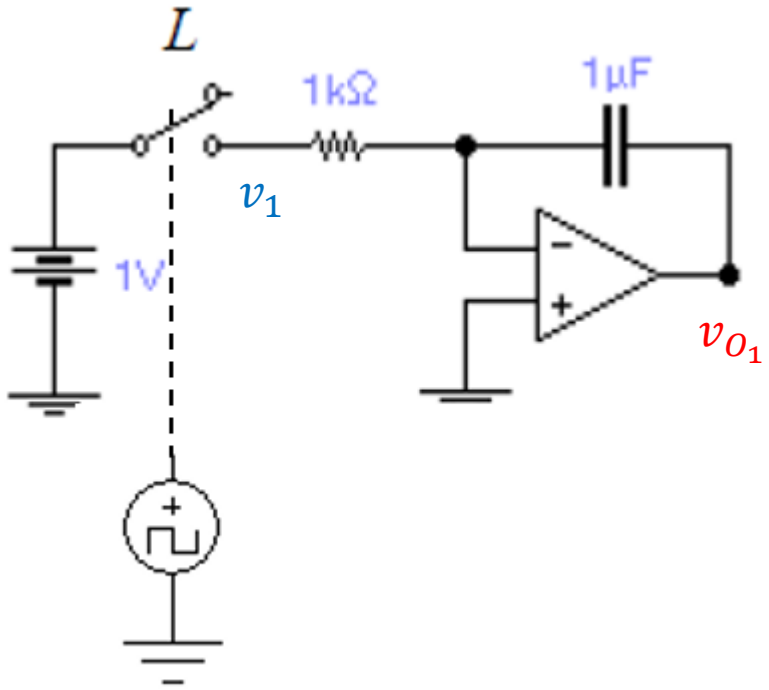
$C = 1\mu F$

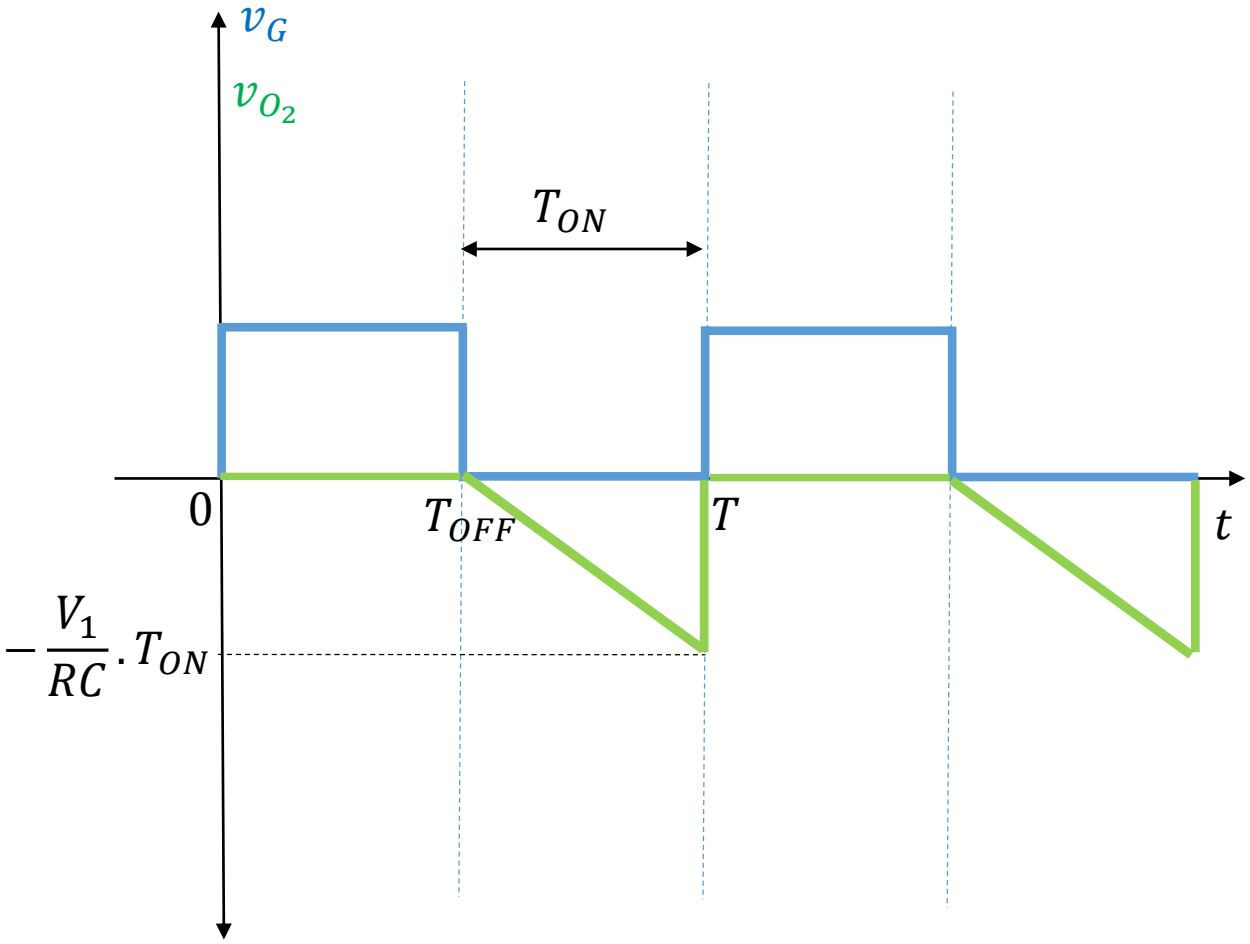


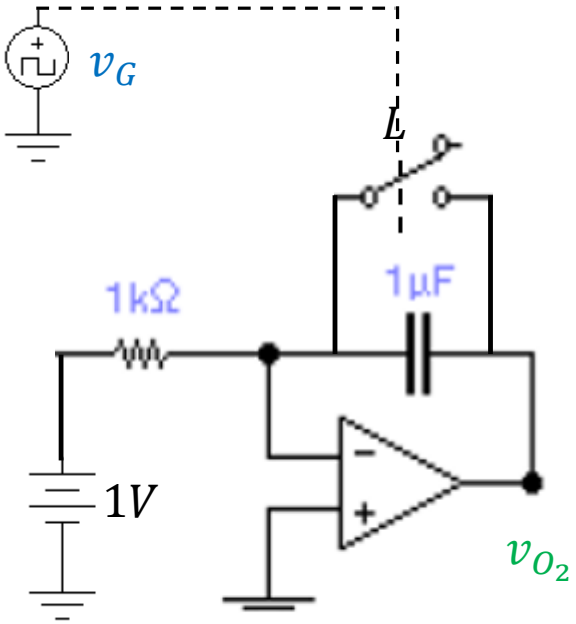
$$i_R = i_C = \frac{v_1}{R}$$

$$v_2 = -v_C = -\frac{1}{C}.\int i_C\ dt\ = -\frac{1}{RC}.\int v_1\ dt = k.\int v_1\ dt \Rightarrow$$ *Amplificador Integrador e Inversor*

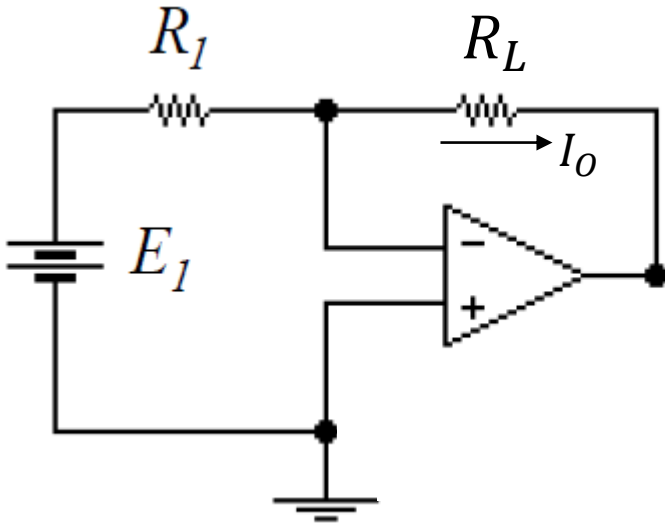

v_1
v_{O_1}
T_{OFF}
$1V$
0
T_{ON}
T
t
$-\frac{V_1}{RC}.T_{ON}$


L
1kΩ
1µF
v_1
1V
v_{O_1}


v_G
v_{O_2}
T_{ON}
0
T_{OFF}
T
t
$-\frac{V_1}{RC}.T_{ON}$

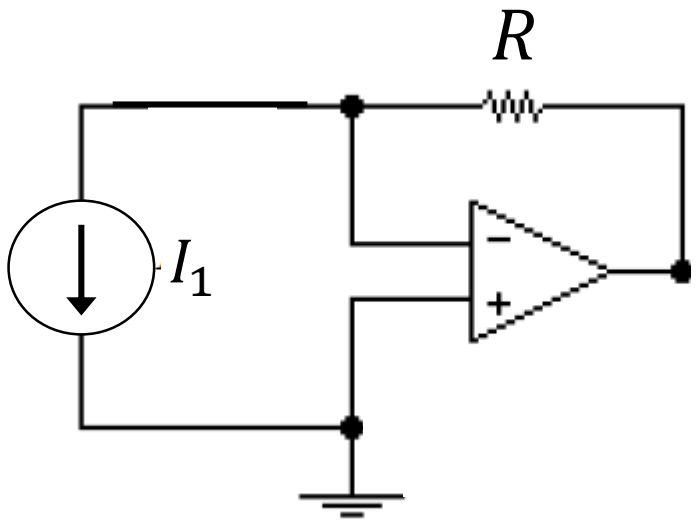

v_G
L
1kΩ
1µF
1V
v_{O_2}

Convertidor Tensión ⟶ Corriente



$$I_O = \frac{E_1}{R_1}$$

Convertidor Corriente ⟶ Tensión



$$E_O = I_1 . R$$